Semantic Hashing: Hashes as Declared Equivalence Classes of Meaning

Hashing as Executable Semantic Collapse Rather Than Byte Integrity

Under this view, hashes do not identify objects or byte sequences, but equivalence classes of meaning defined by an explicit semantic projection (or lens).

Author
Adam Ableman
Constitutional Computing
Contact: ableman.research@gmail.com

Abstract

Hashing is conventionally understood as a cryptographic or engineering primitive: a deterministic function for integrity verification, deduplication, and tamper detection over concrete data. While effective, this framing obscures a deeper and increasingly central systems problem: **how can a system determine that two artifacts are meaningfully the same without re-interpreting, re-reasoning, or trusting narrative explanations?**

This paper introduces **Semantic Hashing**, a reframing of hashing as a *governed semantic operator* rather than a structural checksum. Under this view, hashes do not identify objects or byte sequences, but **equivalence classes of meaning** defined by an explicit semantic projection (or *lens*). Meaning is first declared - by specifying which distinctions are invariant and which are forbidden from relevance - then aggressively normalized and hashed. The resulting hash serves as a stable, addressable identifier for semantic sameness across time, representation, and context.

Central claim. When hashing is applied *after* an explicit, governed semantic collapse, it becomes a mechanism for admissibility, drift detection, and semantic continuity rather than mere integrity checking. Semantic hashing converts value judgments about meaning into enforceable system invariants, enabling refusal without debate, audit without re-interpretation, and continuity without trust.

---

1. The Irreducible Maneuver

At its most reduced form, semantic hashing performs a single maneuver:

1. **Select the meaning you are willing to keep.**
2. **Commit that selection to a canonical form.**
3. **Hash it.**
4. **Let everything else fall away.**

No inference is required. No interpretation is preserved. No appeal is possible after the fact.

What remains is not a summary of the system, but a **decision about what is allowed to survive change**.

1.1 Selection Before Security

The decisive step is not the hash.

It is the moment a system declares:

- *this distinction matters,*
- *this variation does not,*
- *this change is admissible,*
- *this one is not.*

Before any cryptographic commitment occurs, the system must choose what it agrees to remember. Everything not chosen is deliberately abandoned to entropy. It may remain observable, archived, or narratable, but it is denied participation in identity, admissibility, or continuity.

This is not an optimization. It is a value judgment rendered executable.

1.2 Loss as Discipline

The maneuver is intentionally lossy.

Semantic hashing does not attempt to preserve richness, context, or nuance. It replaces them with a hard boundary: *either this survives the maneuver, or it does not.*

This apparent brittleness is the point. Meaning that cannot survive explicit selection cannot be relied upon for governance, admissibility, or continuity. Systems that attempt to carry everything forward ultimately carry nothing with integrity.

Loss, here, is discipline.

1.3 From Brittleness to Instrumentation

In isolation, the maneuver is fragile. A poorly chosen selection collapses too much or too little. A careless projection produces false sameness or spurious difference.

The critical move is not to avoid this fragility, but to **govern it**.

When the process of selection is explicit, versioned, inspectable, and revisable only through declared change, brittleness becomes precision. The system gains the ability to test itself against its own prior commitments and to detect semantic drift without re-interpretation.

1.4 The Hash as a Vow

Only after selection and normalization does hashing occur.

At this point, the hash is no longer a checksum of data. It is a **vow**:

Everything we agreed mattered is captured here. Everything else, we have agreed to forget.

The hash names an equivalence class of meanings. It does not describe them, justify them, or explain them. It simply asserts that anything bearing this hash is indistinguishable under the declared regime of relevance.

1.5 What the Maneuver Enables

This maneuver enables a class of statements most systems cannot make honestly:

- *This is the same rule we enforced before.*
- *This transformation is admissible under the same law.*
- *Nothing we agreed mattered has changed.*

Or, just as importantly:

- *Something that mattered did change.*

Semantic hashing does not make systems correct. It makes them **accountable to their own definitions of meaning**.

---

2. Semantic Hashing and Governed Continuation

Semantic hashing does not stand alone. It composes directly with **Governed Semantic Compilation (GSC)** to form a complete governed continuation protocol.

GSC governs **when an output may be promoted** from proposal to binding state. Semantic hashing governs **what it means for that output to count as the same thing** when promotion is evaluated.

Operationally:

1. Outputs are proposed (GSC preview).
2. A semantic lens extracts invariant structure (semantic hashing).
3. The resulting hash is compared against admissible equivalence classes (GSC validation).
4. Promotion occurs if and only if the hash is admissible; otherwise, refusal is enforced.

This composition defeats authority laundering mechanically. Narrative explanations cannot retroactively assert continuity if the semantic hash has changed.

---

3. Scope and Motivation

Modern systems routinely claim that two things are "the same":

- the same policy,
- the same intent,
- the same rule,
- the same procedure,
- the same outcome under constraint.

These claims are rarely mechanized. Instead, sameness is asserted through narrative, documentation, or institutional memory. When disputes arise, systems fall back to re-interpretation, authority, or re-reasoning - processes that are expensive, ambiguous, and vulnerable to drift.

Hashing is ubiquitous, but almost never used at this semantic level. Hashes protect bytes, blocks, or messages, while meaning is assumed to be stable without being specified. Semantic hashing addresses this gap by making sameness explicit, enforceable, and testable.

---

4. Traditional Hashing and Its Limits

Traditional hashing answers a structural question:

Are these bytes identical?

It excels at integrity, deduplication, and tamper detection. But structural identity is insufficient for semantic identity.

Consider two cases:

- Two policy documents are byte-identical, but one is issued under a superseded authority and the other under a current mandate. A traditional hash declares them the same; a semantic hash incorporating provenance declares them different.
- Two JSON configurations differ only in key ordering. Traditional hashes differ; a semantic hash under a structural-equivalence lens remains stable.

Semantic hashing asks a different question:

Are these meanings equivalent under a declared notion of relevance?

---

5. Declared Meaning and the Semantic Lens

A **semantic lens** is an explicit projection from a rich artifact into a reduced representation containing only invariant-relevant structure.

A minimal characterization treats a semantic lens **L** as a triple:

- **E** : Artifact -> Features (extraction)
- **N** : Features -> Features (normalization)
- **C** : Features -> CanonicalForm (canonicalization)

This decomposition is illustrative, not prescriptive. What matters is not the formalism, but that the projection is explicit, inspectable, and governed.

The lens does not preserve everything. It enforces forgetting. Distinctions excluded by the lens are forbidden from relevance under that regime.

---

6. Targeted Semantic Collapse

Applying a semantic lens produces **targeted semantic collapse**.

Surface variation is destroyed. Incidental structure is erased. Only declared invariants remain. Normalization is intentionally aggressive: sorting unordered sets, canonicalizing identifiers, stripping narrative, collapsing equivalent forms.

After collapse, any remaining difference is semantically meaningful by construction.

---

7. Hashing as Equivalence-Class Commitment

Once collapse is complete, hashing is applied.

The resulting hash does not identify an object. It identifies **membership in an equivalence class** defined by the lens.

All artifacts that project, normalize, and hash identically are semantically identical under that lens. The hash is a name for a semantic region, not a description of a thing.

---

8. Semantic Hashes as Admissibility Tokens

Because semantic hashes identify equivalence classes, they function naturally as admissibility gates.

Admissibility reduces to a simple check:

Does this semantic hash belong to an allowed class?

No re-interpretation is required. If the hash changes, admissibility fails. Refusal becomes mechanical rather than rhetorical.

---

9. Drift Detection Without Re-Reasoning

Semantic hashing enables drift detection without re-evaluation.

If the lens defines what must not change, any semantic drift manifests as a hash change. This exposes silent reinterpretation, gradual erosion, and authority laundering as state changes rather than arguments.

For example:

- Policy v1 -> v2: wording changes, semantic hash stable.
- Policy v2 -> v3: wording unchanged, scope altered, semantic hash changes.

Continuity claims are thus testable, not asserted.

---

10. Temporal Anchoring and Semantic Receipts

A semantic hash anchors meaning in time. It asserts that a particular meaning existed, under a particular lens version, at a particular moment.

Hashes are computed as:

```
H(Lens_ID, Lens_Version, Canonical_Form)
```

Lens evolution is therefore explicit. Old hashes remain valid under old regimes; new hashes signal declared change rather than silent drift.

This yields **semantic receipts**: immutable records that prove what meaning existed when, without freezing future evolution.

---

11. Practical Lens Examples

- **Domain:** Code patches
  Lens Preserves: AST structure, control flow
  Lens Discards: Whitespace, comments
- **Domain:** Legal policy
  Lens Preserves: Jurisdiction, authority, scope
  Lens Discards: Narrative justification
- **Domain:** Function signatures
  Lens Preserves: Arity, types, effects
  Lens Discards: Implementation details
- **Domain:** Workflow definitions
  Lens Preserves: State transitions
  Lens Discards: UI layout, labeling

These are not abstractions; they are explicit answers to the question *"what counts as the same thing here?"*

---

12. Relationship to MCIR

Semantic hashing is a foundational primitive within MCIR-style governed systems.

Laws, transitions, and receipts are each subject to semantic hashing under distinct lenses. Each lens defines what "the same law," "the same move," or "the same history" means. MCIR demonstrates that semantic hashing is operational, not theoretical: it enables admissibility, refusal, and governance without relying on narrative trust or model cognition.

---

13. Failure Modes and Limits

Semantic hashing is not universal.

Failure modes include:

- poorly chosen lenses that collapse too much or too little,
- unacknowledged lens drift,
- governance capture embedding biased invariants,
- adversarial exploitation of weak lenses,
- fragmentation into incompatible semantic regimes.

These failures are visible by design. They surface as explicit changes in declared meaning rather than hidden reinterpretations. Lens contestation and forking are legitimate governance outcomes, not errors.

14. What Semantic Hashing Is Not

Semantic hashing does not determine truth, eliminate disagreement, or capture all nuance. It governs **permission**, not correctness.

Learned or inferred lenses belong to a different regime with different guarantees. Semantic hashing, as defined here, requires lenses to be declared, not discovered.

---

15. Conclusion

Hashing has long protected data. Semantic hashing protects meaning.

By forcing systems to declare what they are willing to treat as invariant - and enforcing that declaration mechanically - semantic hashing converts vague claims of continuity into executable commitments. It allows systems to say, precisely and without rhetoric, *nothing meaningful changed*, or to acknowledge, just as precisely, that it did.

As systems increasingly act on language, policy, and intent, the ability to mechanize semantic sameness without re-interpretation becomes foundational.

The hash is no longer a checksum of bytes. It is a commitment about meaning.

Corpus-Wide Reference Positioning

This paper argues that hashing is conventionally understood as a cryptographic or engineering primitive: a deterministic function for integrity verification, deduplication, and tamper detection over concrete data. Its nearest external comparison fields are crypto, semantic, formal. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: A valid hash, signature, or inclusion proof establishes a bounded cryptographic relation, not meaning, authority, safety, or jurisdiction by itself. Formal vocabularies and graph links make commitments explicit but cannot by themselves establish correct identity, shared interpretation,

or lossless translation between contexts. Formal assurance remains relative to the encoded model, trusted base, and refinement chain; proof does not remove specification error or unmodeled environmental behavior. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] Anders Rundgren, Bret Jordan, and Samuel Erdtman. *JSON Canonicalization Scheme (JCS)*. RFC 8785, 2020. https://www.rfc-editor.org/rfc/rfc8785 **Relationship:** Adjacent canonicalization standard relevant to stable cryptographic commitments.

[**E2**] Ben Laurie et al. *Certificate Transparency Version 2.0*. RFC 9162, 2021. https://www.rfc-editor.org/rfc/rfc9162 **Relationship:** Adjacent append-only Merkle-log standard for inclusion and consistency evidence.

[**E3**] Manu Sporny et al., eds. *Decentralized Identifiers (DIDs) v1.0*. W3C Recommendation, 19 July 2022. https://www.w3.org/TR/did-core/ **Relationship:** Adjacent decentralized identity standard; identifier resolution is not assumed to settle semantic standing or jurisdiction.

[**E4**] Juan Benet. "IPFS - Content Addressed, Versioned, P2P File System." 2014. arXiv:1407.3561. https://arxiv.org/abs/1407.3561 **Relationship:** Predecessor content-addressed network architecture; content identity is compared with, not treated as sufficient for, semantic jurisdiction.

[**E5**] Richard Cyganiak, David Wood, and Markus Lanthaler, eds. *RDF 1.1 Concepts and Abstract Syntax*. W3C Recommendation, 25 February 2014. https://www.w3.org/TR/rdf11-concepts/ **Relationship:** Adjacent graph data model and identity baseline.

[**E6**] Alistair Miles and Sean Bechhofer, eds. *SKOS Simple Knowledge Organization System Reference*. W3C Recommendation, 18 August 2009. https://www.w3.org/TR/skos-reference **Relationship:** Adjacent standard for concepts, labels, schemes, notes, and semantic mappings.

[**E7**] W3C OWL Working Group. *OWL 2 Web Ontology Language Document Overview*, Second Edition. W3C Recommendation, 11 December 2012. https://www.w3.org/TR/owl2-overview/ **Relationship:** Adjacent formal ontology standard used to compare declared semantics and inference commitments.

Ableman corpus

[**A1**] Adam Ableman. *The Three Hash Rule.* Ableman Corpus current edition; work identity `PAPER.PUB.20788617`;
source SHA-256
`F6FA73D083B70711A8C817F7779751D1506FEBA0E356DB9805BB1570663731B5`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Constitutional Semantic Hashing: Governed Commitment Schemes for Semantic Provenance, Admissibility, and Constitutional Continuity.* Ableman Corpus current edition; work identity `PAPER.PUB.20248734`;
source SHA-256
`B1DAC3E2CD4D57BAFBF6462BA6ACFFF7CF120CDB7FA37D0C173525744CC0A2F0`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Cryptographic Admissibility and Jurisdiction.* Ableman Corpus current edition; work identity `PAPER.PUB.20602448`;
source SHA-256
`6094338D8706F63E8787A5E8DCF40F8A651C0FC3E5EA776DD61791566D094D38`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Self-Certifying Proof-Carrying Identity for Registrar-Free Namespace Networks.* Ableman Corpus current edition; work identity `PAPER.PUB.20114888`;
source SHA-256
`1AFDBD390D3BF28224727FE150FFB30CA1810D997778CC8702B6CD20C4FA5B45`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Meaning as Binding.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-5F976917F0B283EA`;
source SHA-256
`A470D1A0B9ED7C3CEAB6BB88BE4B642A2BA36427447F787764329FD574DA0C97`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *Governed Meaning Base (GMB).* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-D95CCC4FB7B58BFB`;

source SHA-256
A3B62E50F8BEE125FDAE0D38C2839AFD1CCC138296268BF93217481E38AA48EA.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *Semantic Manifold Topology.* Ableman Corpus current edition; work identity PAPER.PUB.20257244;
source SHA-256
B8F28CC652A73A3EE4CF6FD0E23C1ABC922A6660319B2FB4CC1CCAAA606EC64A.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
License: CC-BY-ND-4.0

DOI: 10.5281/zenodo.18500533